ABINAYA M



AI/ML Enthusiast |Java Developer
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SUMMARY

Final year Computer Science Engineering Student with focused on Artificial Intelligence & Machine Learning, Java Backend Development. Eager to leverage my programming, problem-solving to develop intelligent applications while continuously learning and contributing to impactful software solutions.

EDUCATION

Bachelor of Engineering in Computer Science & Engineering **2023 -2027**
Kongunadu College of Engineering and Technology, Trichy CGPA: 7.76/10.0

Higher Secondary Class XII **2022-2023**
Aringar Anna Goverment Higher Secondary School, Pachaperumalpatty, Trichy. Percentage : 76 %

PROJECTS

AI-Powered User Monitoring & Anomaly Detection System

Technologies Used: Python, Flask, Scikit-learn, MongoDB Atlas, psutil, Isolation Forest, SMTP, HTML, CSS.

- Developed an AI-driven anomaly detection system using Isolation Forest algorithm to identify unusual user system behavior.
- Executed real-time monitoring of CPU, memory, disk usage, and user activities using Python and psutil.
- Integrated automated email alerts using SMTP to notify administrators of suspicious activities, enabling system monitoring.

Flood Mapping from Satellite Imagery using Deep Learning

Technologies Used: Python, PyTorch, UNet++, ResNet-34, Rasterio, Albumentations, NumPy, Pandas.

- Enhanced an end-to-end deep learning pipeline to detect and segment flood-affected regions from satellite imagery using UNet++ with a ResNet-34 encoder.
- Performed data preprocessing, augmentation and geospatial image handling using Rasterio and Albumentations.
- Implementing the complete machine learning workflow during the AISEHack – ANRF Flood Mapping Challenge.

Enhanced Library Management System

Technologies Used: Java, Spring Boot, MySQL, HTML, CSS, JavaScript , JWT Authentication

- Enhanced a web-based Library Management System to automate book management, borrowing, and return processes.
- Applied secure role-based access control using Spring Boot and JWT authentication.
- Merged features such as Smart Book Recommendations, Automated Fine Calculation, and User Management.

TECHNICAL SKILLS

Languages	**:** Java, Python, SQL.
Web Technologies	**:** HTML5, CSS3, JavaScript, JWT.
AI/ML	**:** Isolation Forest, UNet, ResNet, Rasterio, Albumentations.
Databases	**:** MySQL, MongoDB.
Libraries & Frameworks	**:** Scikit-learn, Spring Boot, NumPy, Pandas, psutil.
Core Concepts	**:** OOP, DBMS, REST APIs.

EXPERIENCE

Frontend Development Intern **Aug to Sep 2025**
IBM SkillsBuild | Remote

- Completed comprehensive frontend development training program with hands-on projects.
- Designed responsive web interfaces using HTML, CSS, Java Script, Successfully completed.

HACKATHON PARTICIPATION

- **CMR National Level Hackathon 2024 – Hyderabad** Participated as Team Lead and developed a cybersecurity anomaly detection system.
- **ANRF – AISEHack 2026 - Flood Detection (IBM and IIT Delhi at IIIT Hyderabad**) Achieved a Top 10 the AISE–ANRF Flood Mapping Challenge, Built an AI-based flood detection system using satellite imagery to identify and map flooded regions accurately**.**

CERTIFICATIONS

- **AWS Cloud Practitioner Essentials |** AWS Training & Certification July 2025
- **Python Programming** |Infosys Springboard May 2024